UltraGroth. Lookup Checks Enabled in Groth16

October 29, 2026

Blockstream Research

research.blockstream.com

github.com/BlockstreamResearch/zk-book



Plan

1 Introduction

2 Groth16 Recap

3 UltraGroth

Introduction

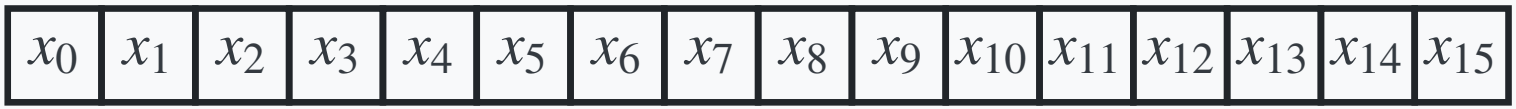
Motivation

We typically want to check **inclusion** $\{z_i\}_{i\in[n]} \subseteq \{t_j\}_{j\in[v]}$, where $\{z_i\}_{i\in[n]}$ is part of the witness while $\{t_j\}_{j\in[v]}$ is the lookup table.

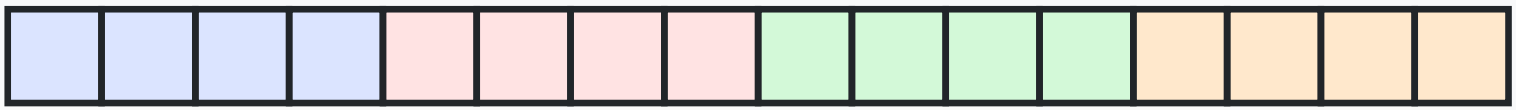
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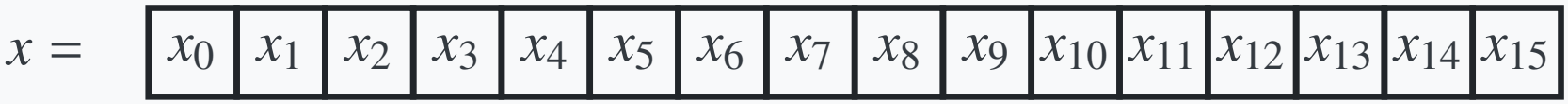
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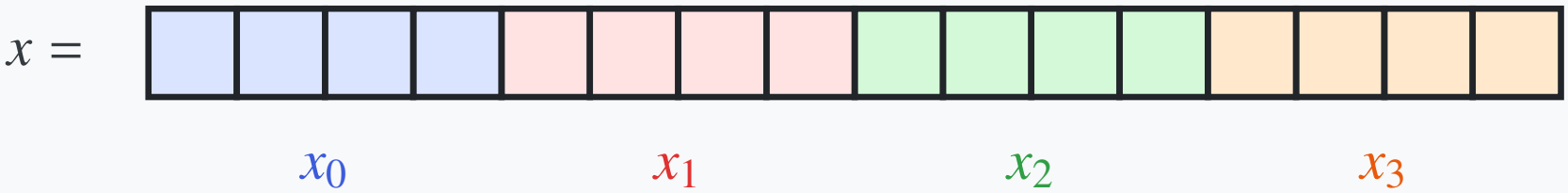
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Generally: for n-bit range-check, the circuit's complexity reduces from $O(n)$ to $O(2^w + \frac{n}{w})$, which yields $O(n/\log n)$ asymptotic.

Logup Check

Theorem (On fractional lookup check)

The inclusion check $\{z_i\}_{i\in[n]} \subseteq \{t_i\}_{i\in[v]}$ is satisfied if and only if there exists the set of multiplicities $\{\mu_i\}_{i\in[v]}$ where $\mu_i = \#\{j \in [n] : z_j = t_i\}$ such that for $\gamma \leftarrow\$ \mathbb{F}$:

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Problem: We *cannot define random* signals in Circom since Groth16, compared to $\mathcal{P}$lon$\mathcal{K}$ or sumcheck-based approaches, is not compiled from interactive protocol using Fiat-Shamir heuristic.

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$$C_{i,j} = \sum_{k \in [n]} A_{i,k} B_{k,j} \qquad \textit{// Costs } n \textit{ constraints per } C_{i,j}$$

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We can instead apply the **Freivalds' protocol**. Sample random $\boldsymbol{\gamma} \leftarrow\!\!\$\ \mathbb{F}^n$, compute C off-circuit and then verify:

$$AB\boldsymbol{\gamma} = C\boldsymbol{\gamma} \qquad \textit{// Costs } 3n^2 \textit{ constraints}$$

Example: Attention layer implementation in zkML.

Roadmap

1. We recap the Groth16 construction.
2. We identify how to make it interactive.
3. We specify how Fiat-Shamir transformation should be applied.
4. We show how it can be practically implemented.

Groth16 Recap

R1CS

Recall that we can encode any NP statement in the form of m equations of form $\langle \boldsymbol{\ell}_j, \mathbf{z} \rangle \cdot \langle \boldsymbol{r}_j, \mathbf{z} \rangle = \langle \boldsymbol{o}_j, \mathbf{z} \rangle$ for $j \in [m]$ and $\mathbf{z} \in \mathbb{F}^n$. Such way of representing the statement is called **R1CS arithmetization**.

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- ✓ As of now, this is one of the most optimal arithmetization systems available (compared to PlonK and AIR).
- ✓ All linear operations over elements of $\mathbf{z}$ cost **0 constraints**, compared to PlonK.

R1CS: Example

Example

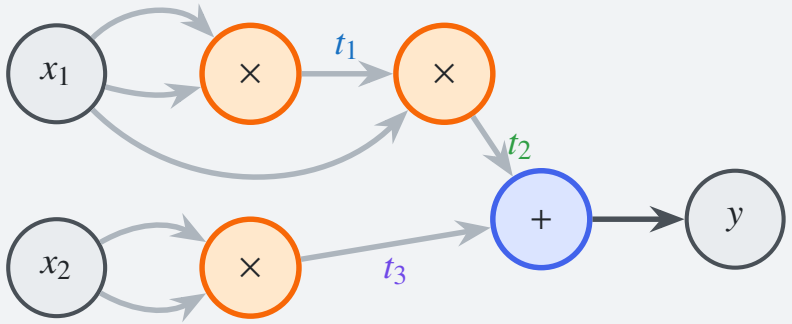
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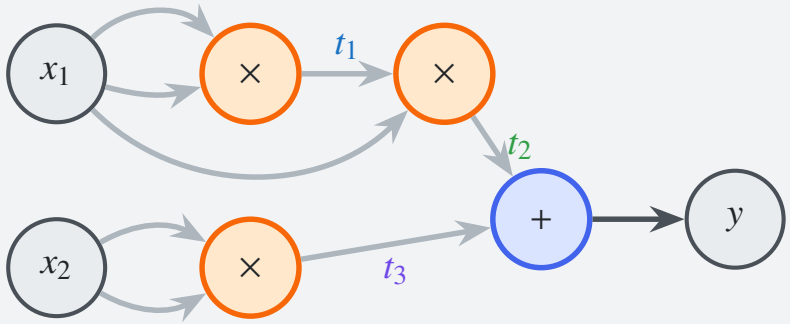
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In this case, the witness looks as $\mathbf{z} = (1, x_1, x_2, t_1, t_2, t_3, y)$, and (for simplicity, consider only L, R):

$$L = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

QAP

We interpolate the columns of each of the matrices, thus getting $3n$ polynomials $\{(\ell_i(T), r_i(T), o_i(T))\}_{i\in[n]} \subseteq \mathbb{F}[T]^{\leq m}$:

$$\ell_i(\omega^j) = L_{j,i}, \quad r_i(\omega^j) = R_{j,i}, \quad o_i(\omega^j) = O_{j,i}, \quad i \in [n],\ j \in [m]$$

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$$\sum_{i\in[n]} z_i \ell_i(T) \cdot \sum_{i\in[n]} z_i r_i(T) = \sum_{i\in[n]} z_i o_i(T) + t_\Omega(T) h(T),$$

where $h(T)$ is computed by a prover and $t_\Omega(T) \triangleq \prod_{h\in\Omega}(T - h)$ is the *vanishing polynomial* over *evaluation domain* $\Omega = \{\omega^j\}_{j\in[m]}$. The corresponding relation:

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$$\mathcal{R}_{\mathsf{QAP}} = \left\{ \begin{matrix} \mathbb{x} = \{z_i\}_{i\in\mathcal{I}_X} \\ \mathbb{w} = \{z_i\}_{i\in\mathcal{I}_W} \end{matrix} \;\middle|\; \begin{matrix} \sum_{i\in[n]} z_i\ell_i(T) \cdot \sum_{i\in[n]} z_i r_i(T) = \sum_{i\in[n]} z_i o_i(T) + t_\Omega(T)h(T) \\ \text{for some } h(T) \in \mathbb{F}[T] \end{matrix} \right\}$$

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- Verify$(\boldsymbol{\sigma}, \mathbb{x}, \pi) \to \{0, 1\}$. The verifier gets the arithmetic circuit $t : \mathbb{F}^{m+k} \to \mathbb{F}^\eta$ of degree d and verifies whether $t(\boldsymbol{\sigma}, \boldsymbol{\pi}) = 0$.

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Groth16 is essentially a Linear NIP where $d = 2$ and $\boldsymbol{\sigma}$ is given by:

$$\boldsymbol{\sigma} = \left(\alpha, \beta, \gamma, \delta, \{\tau^i\}_{i \in [n]}, \left\{ \frac{f_i(\tau)}{\gamma} \right\}_{i \in [m]}, \left\{ \frac{\tau^i t_\Omega(\tau)}{\delta} \right\}_{i \in [n]} \right),$$

with $f_i(T) := \beta \ell_i(T) + \alpha r_i(T) + o_i(T)$ and $\boldsymbol{\tau} = (\alpha, \beta, \gamma, \delta, \tau)$.

Groth16 Construction

Fix bilinear group $\mathcal{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e)$ with pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$.

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$$a(T) = \alpha + \sum_{i \in [n]} z_i \ell_i(T) + r\delta,\ b(T) = \beta + \sum_{i \in [n]} z_i r_i(T) + s\delta,$$

$$c(T) = \delta^{-1} \left(\sum_{i \in \mathcal{I}_W} z_i f_i(T) + h(T) t_\Omega(T) \right) + a(T)s + b(T)r - rs\delta$$

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- Verify$(\mathsf{vp}, \mathbb{x}, \pi) \to \{0, 1\}$. Parse $\pi = (\pi_A, \pi_C, \pi_B)$ and accept the proof if and only if

$$e(\pi_A, \pi_B) = e(g_1^\alpha, g_2^\beta) \cdot e(g_1^{\iota(\tau)}, g_2^\gamma) \cdot e(\pi_C, g_2^\delta),$$

where $\iota(T) := \gamma^{-1} \sum_{i\in\mathcal{I}_X} z_i f_i(T)$ is the input commitment.

UltraGroth

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Problem. We cannot practically "hash" the witness part $\mathbb{w}_0$.

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Round 1: $\mathcal{P}$ samples the challenge $\mathbb{x}_1 \leftarrow \mathsf{H}(\mathsf{pp}, \pi_C^{\langle 0 \rangle})$, samples $r, s \leftarrow\$ \mathbb{F}$ and computes $\pi_C^{\langle 1 \rangle} \leftarrow g_1^{c_1(\tau)}$ as:

$$c_1(T) = \delta^{-1} \left(\sum_{j \in \mathcal{I}_W^{\langle 1 \rangle}} z_j f_j(T) + h(T) t_\Omega(T) \right) + a(T)s + b(T)r - r_0 \delta_0 - rs\delta$$

2-round UltraGroth Construction: The rest

Then, parts $\pi_A \leftarrow g_1^{a(\tau)}$ and $\pi_B \leftarrow g_2^{b(\tau)}$ are computed as usual via:

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where $\iota(T) = \gamma^{-1} \sum_{i \in \mathcal{I}_X} z_i f_i(T)$ as before and $\mathbb{x}_1 = \mathsf{H}(\mathsf{pp}, \pi_C^{\langle 0 \rangle})$.

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Conclusion

UltraGroth protocol's verifier is only **4 pairings**, **1 hashing operation**, and $O(|\mathbb{x}|)$ exponentiations over $\mathbb{G}_1$.

Multi-round UltraGroth

Definition (dQAP)

We define the $(d+1)$**-round quadratic arithmetic program** (or d**QAP**, for short), as follows:

$$\mathcal{R}_{\mathsf{dQAP}} = \left\{ \begin{array}{l} \mathbb{x}_i = \{z_j\}_{j \in \mathcal{I}_X^{\langle i \rangle}} \\ \mathbb{w}_i = \{z_j\}_{j \in \mathcal{I}_W^{\langle i \rangle}} \\ \text{for } i \in [d+1] \end{array} \middle| \begin{array}{c} \ell(T) \cdot r(T) = o(T) + t_\Omega(T) h(T) \\ \ell(T) = \sum_{i \in [n]} z_i \ell_i(T), \\ r(T) = \sum_{i \in [n]} z_i r_i(T), \\ o(T) = \sum_{i \in [n]} z_i o_i(T), \\ \text{for some } h(T) \in \mathbb{F}[T] \end{array} \right\},$$

where $\{\mathcal{I}_X^{\langle i \rangle}\}_{i \in [d+1]}$ and $\{\mathcal{I}_W^{\langle i \rangle}\}_{i \in [d+1]}$ partition $[n]$.

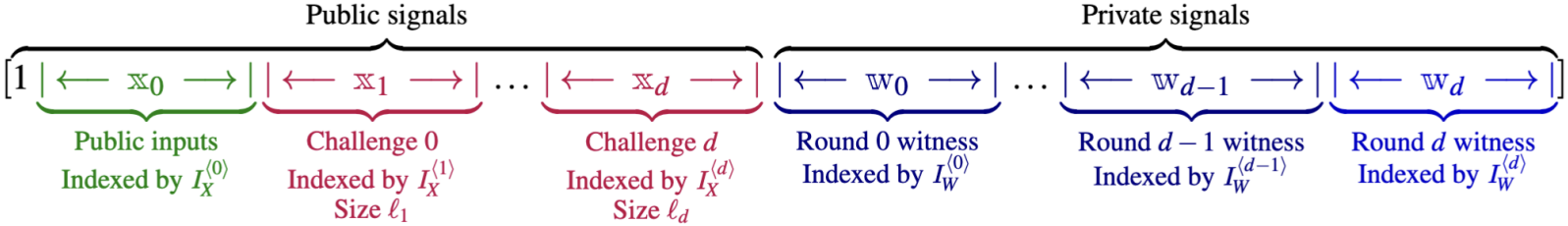
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Define **strategy** for $\mathcal{R}_{\mathsf{dQAP}}$ as the collection of functions $S = \{S_i\}_{i\in[d]}$ each of which computes the witness for the given round given previous witnesses and challenges and the current challenge, sampled by the verifier. In other words,

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Example

0QAP represents the regular QAP with the strategy $S = \{S_0\}$ that consists of the witness generator: $\mathbb{w} = S_0(\mathbb{x})$.In turn, 1QAP represents the lookup Groth16 version where $\mathbb{w}_0 = S_0(\mathbb{x}_0)$ computes the witness without lookups while $\mathbb{w}_1 = S_1(\mathbb{x}_0, \mathbb{x}_1, \mathbb{w}_0)$ computes lookup constraints.

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- If $i < d$, for each $j \in \mathcal{I}_X^{\langle i+1 \rangle}$, set $z_j \leftarrow$ H(a_{i+1}, g_1^j).

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Verification: $e(\pi_A, \pi_B) = e(g_1^\alpha, g_2^\beta) \cdot e(g_1^{\iota(\tau)}, g_2^\gamma) \cdot \prod_{i \in [d+1]} e(\pi_C^{\langle i \rangle}, g_2^{\delta_i})$.

UltraGroth Efficiency

Groth16 performance over the circuit of size n and statement size ℓ.

- **Prover work**: MSM of size $O(n)$ over $\mathbb{G}_1$ and $\mathbb{G}_2$.
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UltraGroth performance over $\mathcal{R}_{\mathsf{dQAP}}$ in turn:

- **Prover work**: MSM of size $O(n/\log n)$ over $\mathbb{G}_1$ and $\mathbb{G}_2$.
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- **Verifier work:** $(d+3)$ pairings + $O(\ell)$ $\mathbb{G}_1$ exps + $\sum_{i\in[d+1]\setminus\{0\}} |\mathbb{x}_i|$ hashing operations.
- Allowed interactiveness for potentially more complex protocols.

Thank you for your attention



research.blockstream.com

github.com/BlockstreamResearch/zk-book



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